

Some Macroeconomic Implications of an Information Age*

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Abstract

We study the macroeconomic implications of three key features of an *information age*: i) the information technology revolution favors information production and hence information compared to other inputs of production becomes relatively cheaper; ii) information is an intermediate input that, once produced, can be reused at no extra cost (the marginal cost of using information is almost zero); iii) information production is human capital intensive. These features are incorporated into a dynamic general equilibrium model and the implications of an information revolution for productivity, distribution and welfare are analysed.

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1 Introduction

The past two decades have witnessed extraordinary changes in the way we work, live, and communicate with one another. The spread of information and knowledge now occurs at speeds that would have been unthinkable only thirty years ago. In the past decade, these changes were accompanied by strong economic growth and rising productivity. One view of these developments holds that we are in the midst of a new industrial revolution driven by information technology and that in this “new economy” the conventional wisdom of the old bricks and mortar economy no longer applies. A more modest view is simply that technology has changed in important ways and all that is needed is an analysis of how these changes in technology affect pricing, markets, and productivity. A more skeptical view is that the information technology boom does not even rank with the great innovations of the twentieth century.¹ Our goal in this paper is to analyze explicitly and formally how an information technology revolution affects an economy, with a particular emphasis on how it changes productivity, income distribution, and economic welfare.

To make progress on this question one needs to understand the role of information and its availability in the economy. We approach this issue by treating information as a central attribute of goods. The key feature of information is that, once it is produced, it can be used repeatedly without much additional cost - the marginal cost of using information after it is produced is almost zero. Software that costs hundreds of millions of dollars to develop can be copied on a CD for a few cents, 100-million dollar movies can be copied on a videotape for a few dollars. Similarly, a 100-million dollar sports event —an NBA final or a soccer match— can be enjoyed by an additional sports fan with almost no extra cost. Tens of millions of dollars are spent for the development and maintenance of professional corporate Web sites through which an additional customer can be served at zero marginal cost. The cost of producing the first copy of an information-intensive good is often substantial, but the cost of producing (or reproducing) additional copies is negligible. These features

¹See for example Frances Cairncross (2000) for a very rosy view of the “new economy”, Shapiro and Varian (1999), for an example of the more modest view and Robert Gordon (2000), for the most skeptical view.

of information-intensive goods have been stressed by other authors.

The production of information itself is human capital intensive. This means that, in an information age, human capital will become a more important determinant of economic success. Human capital investment decisions play an important role in the nature of the goods that are produced and have important implications for the distribution of income.

To capture these ideas we proceed by characterizing all goods in terms of their information content. On this view, information is an input to the production process. We can think of goods as being produced with two types of intermediate input: information inputs and non-information inputs. The criteria for classifying inputs is the marginal cost of usage: the marginal cost of information inputs is zero while the marginal cost of non-information inputs is always non-zero. This framework implies a very broad notion of information. Essentially, anything that, *once produced*, can be *reproduced costlessly* is information. For our purposes books, databases, software, magazines, music, stock quotes, Web pages, scientific knowledge, are all information. It is obvious that in an information age, information will become more important than other “bricks-and-mortar” type inputs, and will be a larger part of the production of consumption goods. Moreover, the nature of the goods themselves change. Consider the following anecdote: In the late 1980’s at the peak of the holiday season United Parcel Service would handle 3 million inquiries (phone calls) about shipments with an average cost of \$2 per call. In 2001 at the peak of the holiday season they handled more than 9 million inquiries a day at a cost of a few cents per inquiry. UPS now views its product as both the service of transporting a package as well as the information about the location and progress of a shipment.

Given this notion of information, we can characterize all products in terms of their information intensity. Consider, for instance, the production process for delivering knowledge by teaching. Teaching involves non-information inputs like buildings and equipment that need to be provided to a marginal student at a positive —probably, at the optimal size, constant or increasing— marginal cost. The teaching activity itself, though, is an information input which is performed once regardless of the number of students in a classroom or on a network. When the shares of information and non-

information inputs in the total cost of teaching is considered, the share of information inputs (cost of teachers) outweighs the share of non-information inputs, i.e. teaching is information intensive. In an information age, when access to students is less limited by the need for physical inputs, the information intensity of teaching increases. In contrast, consider more traditional bricks-and-mortar manufacturing goods like consumer durables and producer durables. For these goods a larger portion of total inputs are likely to be non-information intermediate inputs so that manufacturing is a less information intensive production process.

In subsequent sections we analyze some of the macroeconomic implications of an information revolution. In particular we illustrate the consequences of an increase in the information intensity of production, given the main feature of information that, once it is produced, it can be reproduced costlessly. We begin by offering a precise definition of information goods. We proceed to define a general equilibrium environment in which agents choose the human capital investments that will determine the kinds of goods they produce and have incentives to create new goods. We then show how an economy changes with an information revolution that increases the efficiency of information production and distribution.

We find that an information revolution, has important implications for the evolution of the economy and that these changes play out over many decades. In particular, an information age will lead to an increase in income inequality, a long fall and then a sharp rise in measured productivity, an increase in concentration, sharp declines in the prices of high-information goods, a fall and a subsequent increase in the value of the stock market, and an increase in the speed of diffusion of new products to households. All of these features are consistent with observations over the past two and half decades.

2 The Economy

2.1 Information Goods

We begin by assuming there are two types of goods defined on the set $G = \{\underline{\theta}, \bar{\theta}\}$, $0 < \underline{\theta} < \bar{\theta} < 1$.² One unit of a type θ good can be produced according to the technology

$$g(\theta) = \{(i, m) \mid i^\theta m^{1-\theta} = 1\}, \quad \theta \in G, \quad (1)$$

where i is the information input and m is the non-information input used in producing this good. Here $g(\theta)$ is the set denoting all possible (i, m) input couples which produce one unit of a good of type θ . As θ increases the good becomes more information intensive. We assume that all goods require both non-information and information inputs. The intermediate information input is produced at time t via the production function

$$i = z\gamma^t h,$$

where $z > 0$ and $\gamma > 1$ are productivity coefficients and h is the human capital of the producer. The non-information intermediate input is produced according to the production function

$$m = \gamma^t n,$$

where n is the amount of labor used.

2.2 Agents

Consider an infinite-horizon continuous time economy. Each agent is born with a fixed ability level, h , which is distributed over the population according to the cumulative distribution function $\Lambda(h)$.

This is an economy in which agents have a taste for variety; the ability to consume new types of goods gives them utility. For simplicity, we assume symmetry across different types of goods in terms of the utility that they bring to the agents. Thus, agents value different types of goods the same way, i.e., they do not prefer more

²We consider the case of two types of goods, high-information and low-information, for simplicity. The extension to a continuum is straightforward.

information intensive goods to the less, per se, and vice versa. Let the total utility the agent receives be denoted by $U(\mathbf{c})$,

$$U(\mathbf{c}) = \log \mathbf{c}$$

with $\mathbf{c} = \left[\int_0^N c_i^\phi di \right]^\frac{1}{\phi}$, $0 < \phi < 1$, where c_i denotes agent's consumption of good i , N is the number of goods that the household consumes.

Each period an agent has two work options. He can either work as an R&D worker creating new goods or he can directly produce a good. If he works as an R&D worker he earns income w_h , while if he decides to produce a good he needs to choose the optimal type of good (θ and τ) and the optimal non-information and information inputs (m, i). Let $\bar{w}_h$ denote the income an agent with human capital level h earns. Then the agent's work decision is summarized by

$$\bar{w}_h = \max \left[\underbrace{w_h}_{\text{R\&D}}, \max_{\tau, \theta, (m, i), \underbrace{[(p_\tau(\theta) - f_\tau(\theta)) y]}_{\text{production}}} \right],$$

s.t.

$$i^\theta m^{1-\theta} = 1, \tag{2}$$

$$i = z\gamma^t h, \text{ and } m = \gamma^t n,^3$$

$$ny = 1. \tag{3}$$

On the production side, depending on her human capital, the agent decides which type of good to produce, that is, she chooses a θ and τ . Having decided on θ and τ , the agent chooses the optimal (i, m) couple. Finally, given the market price of her product $p_\tau(\theta)$ and $f_\tau(\theta)$ which represents a licensing fee that must be paid to the innovator of a type θ product of age τ , she decides the quantity of output to produce.⁴ Equation (2) ensures that $(i, m) \in g(\theta, j)$. Finally, (3) is the agent's time constraint.

The consumption decision of an agent is distinct from his production decision. Consider an agent with human capital level h . This agent seeks to maximize lifetime

³Time subscripts are omitted in other equations for simplicity of notation.

⁴We describe the determination of this licensing fee further below.

utility by picking optimal consumption, i.e., at time zero his problem is

$$\max_{\{\mathbf{c}_t\}_0^\infty} \int_0^\infty U(\mathbf{c}_t) e^{-\delta t} dt, \quad \delta > 0, \quad (P1)$$

s.t.

$$\int_0^{N_t} p_{it} c_{it} di + \frac{da_t}{dt} = \bar{w}_h, \quad (4)$$

$$c_i \in \{0, [\underline{c}, \infty)\} \text{ for all } i, \quad (5)$$

where p_{it} is the price of the i th good, a_t is the households' assets.

Notice that in (3) an agent has to spend n units of time to produce the non-information input for each unit of the good y that he produces whereas he does not need to spend any time at all to produce the information input. The amount of information input he produces depends on his human capital and he has to produce the information input only once regardless of the amount of output, y . This is the crucial difference between the information and non-information input.

On the consumption side each agent decides how much of each product to consume. The budget constraint, equation (4) says that consumption plus assets carried into the next period must equal the return on assets from the previous period plus the return from producing output y . That return is simply the revenue from selling y , minus the cost of the licensing fee. Because of the form of the utility function, every agent would like to consume some of each variety of goods. Here we assume that there must be at least a minimum level of consumption of each good, $\underline{c}$, due to the constraint (5). This means that not all agents will be able to consume all goods.

The state of an agent in this economy will be given by $s = (h, a)$, the agent's human and physical capital h , and a respectively. Let the measure of agents with state s be denoted by $\phi(s)$. Consider an age- j agent with state $s = (\lambda, a)$. Let his best product type be θ . Notice that the product-type decision—which type of product to produce—is a static decision. Given the human capital of the individual he just picks the type and the age of good, θ , and τ that maximize his current net profit,

$$\max_{\theta, \tau} [p_\tau(\theta) - f_\tau(\theta)] y_\tau(\theta) \quad (6)$$

where $p_\tau(\theta)$ and $f_\tau(\theta)$ are given. Here the amount of type θ good that the agent can produce, $y_\tau(\theta)$, depends on his human capital.

3 New Product Creation

In this section we describe the decision to create new products. We also describe how the costs of creating new products is covered. Here we make the relatively simple assumption that the right to produce all new innovations is licensed and that imitation is also possible.⁵ New products can be developed by incurring a product development cost. This fixed cost includes cost of product specification, pilot plant and prototype, plant and equipment, and manufacturing and marketing start-up as well as the innovation cost that goes for direct research and development. Assume that an amount $\frac{\omega_\theta}{\gamma^t}$ of the composite labor input (i.e., $\lambda(h)$ units of labor-time from each ability level) is necessary for product development⁶. This means that the product development cost of $\square_\theta = \frac{\omega_\theta}{\gamma^t} \int_h w_h \lambda(h) dh$ is incurred for every new product developed. Let the number of agents producing a type θ good be denoted by η_θ . Let the number of products of type θ be denoted by N_θ . Let's assume that information intensive goods require more fixed (R&D) costs, $\omega_{\bar{\theta}} > \omega_{\underline{\theta}}$.

Imitating an existing product is also possible, but only if a fixed cost is paid. The fixed cost of imitating a product at age s is

$$C(s) = \left[\lambda e^{-\rho s} + (1 - \lambda) \right] \square_\theta, \quad s \geq 0, \quad \rho > 0,$$

where s is the time after innovation of the product. The imitation cost has two components. An imitator usually spends much less time and money on research than the innovator because the product's existence and characteristics provide with a great deal of information that the innovator had to obtain through his own costly research. Let's assume that λ is the fraction of the innovation cost that goes to research that the imitator can benefit from. The time it takes to imitate a new product can generally be reduced by spending more money. Each product's imitator is confronted by a time-cost trade-off function, which is the relationship between the amount spent by the imitator and the length of time it would take to imitate this new product. Therefore,

⁵We stay away from discussions of patents and patent policy but we take these issues up in a related paper.

⁶Assuming R&D activity to be more human capital intensive could be more realistic. However, here, we want to exclude increasing inequality due to increased R&D to be able to focus on the "star economy" feature of the model

the cost of imitation decays exponentially at rate ρ after innovation, i.e. imitating a product becomes easier as the product gets older. On the other hand, an imitator often has to go through the same steps as the innovator with respect to pilot plant or prototype construction, investment in plant and equipment, training the employees, and manufacturing and marketing start-up. A $(1 - \lambda)$ fraction of innovation cost represents those kind of costs which do not decay with time. In our taxonomy, high λ goods are high information content goods.

Producers can also produce a certain product by purchasing the license to produce the good from the innovator at each point in time. The decision problem for the innovator is how many licenses to issue at each point in time. Given the quantity of licenses the price of a license is determined in the market. If the innovator issues too few licenses the price of a license will become high enough to attract imitators of the product. Hence innovators are disciplined by the possibility of current and future imitations. Let $f_{\tau,\theta}(q)$ be the equilibrium price of a unit license for a type θ product given the quantity available q . We can now describe the timing of the Cournot game played by the inventor and possible imitators. After paying the invention cost $\square_\theta = \frac{\omega_\theta}{\gamma^t} \int_h w_h \lambda(h) dh$ the innovator decides the number of licenses she will issue for the period. Let's consider the problem of an innovator of a product of age s . Let m_s ⁷ be the number of producers of the innovators's product including the innovator himself (number of imitators plus one). Then the problem of an innovating firm with product at age s at time zero will read

$$V^o(s) = \max_{\{q_{s+t}^o\}_{t=0}^\infty} \int_0^\infty f_{s,\theta}(\mathbf{Q}_{s+t}) q_{s+t}^o e^{-\delta t} dt,$$

where $\mathbf{Q}_{s+t} = [q_{s+t}^o, q_{s+t}^1, q_{s+t}^2, \dots, q_{s+t}^{m_{s+t}}]$ and $f_{s,\theta}(\mathbf{Q}_{s+t})$ is the equilibrium license fee for the product given the quantity of licences issued; q_{s+t}^o is the quantity of licences that the innovator issues, and q_{s+t}^i is the quantity that the i th imitator issues.

There is symmetry across producers (both the innovator and the imitators). Let t_i denote the time difference between dates at which the innovator and the i th imitator start producing. In equilibrium, due to free entry no innovator and imitator should

⁷From here on, we omit the time subscript for simplicity of notation. Also, in some places current time is taken as time 0 for convenience of notation.

make any profits. Consider the value of an innovating firm at the date of innovation, $V_t^o(0)$. Since the first imitator imitates t_1 units of time after innovation date, the innovator produces the product alone as a monopolist between time t and $t + t_1$. Let Π_{t+j}^o , $0 \leq j \leq t_1$ denote the profit stream of the innovator between time t and $t + t_1$. Since there won't be any difference between the innovating firm and any imitator, the value of the imitating firm when the first imitator imitates should be equal to the value of the first imitating firm at that date. Therefore,

$$V_t^o(0) = \int_0^{t_1} \Pi_t^o e^{-\delta t} dt + e^{-\delta t_1} V_{t+t_1}^1(t_1).$$

Similarly, between time $t + t_1$ and $t + t_2$ the innovating firm and the first imitator produce in the market together. Let Π_{t+y}^1 , $t_1 \leq y \leq t_2$ denote the profit stream of these firms during that time. Hence,

$$V_{t+t_1}^1(t_1) = \int_{t_1}^{t_2} \Pi_{t+y}^1 e^{-\delta(y-t_1)} dy + e^{-\delta(t_2-t_1)} V_{t+t_2}^2(t_2).$$

In general, the value of the i th imitator will be given by

$$V_{t+t_i}^i(t_i) = \int_{t_i}^{t_{i+1}} \Pi_{t+y}^i e^{-\delta(y-t_i)} dy + e^{-\delta(t_{i+1}-t_i)} V_{t+t_{i+1}}^{i+1}(t_{i+1}).$$

In equilibrium, if there is product innovation, the cost of innovation has to be equal to the value of an innovating firm, i.e.,

$$\square_\theta = V_t^o(0).$$

Also in equilibrium if i th imitator of a product imitates at product age s then the cost of imitation has to be equal to the value of imitation, i.e.,

$$C(s) = [\lambda e^{-\rho s} + (1 - \lambda)] \square_\theta = V^i(s),$$

Using these equilibrium conditions, it follows that, in equilibrium,

$$\begin{aligned} \int_0^{t_1} \Pi_t^o e^{-\delta t} dt &= \square_\theta - e^{-\delta t_1} C(t_1), \\ &= \square_\theta - e^{-\delta t_1} [\lambda \square_\theta e^{-\rho t_1} + (1 - \lambda) \square_\theta], \\ &= \square_\theta (1 - \lambda e^{-(\rho+\delta)t_1} - (1 - \lambda) e^{-\delta t_1}), \end{aligned}$$

and

$$\begin{aligned}
\int_{t_1}^{t_2} \Pi_{t+y}^1 e^{-\delta(y-t_1)} dy &= C(t_1) - e^{-\delta(t_2-t_1)} C(t_2), \\
&= \left[\lambda \square_\theta e^{-\rho t_1} + (1-\lambda) \square_\theta \right] - e^{-\delta(t_2-t_1)} \left[\lambda \square_\theta e^{-\rho t_2} + (1-\lambda) \square_\theta \right], \\
&= \square_\theta \left(\lambda e^{-\rho t_1} + (1-\lambda) - \lambda e^{-(\rho+\delta)t_2+\delta t_1} - (1-\lambda) e^{-\delta(t_2-t_1)} \right),
\end{aligned}$$

which can be carried on recursively⁸. Let m^o denote the eventual number of imitators for a product. The m^o th imitator should also be making zero profits in equilibrium. Therefore,

$$\int_{t_{m^o}}^{\infty} \Pi_{t+y}^{m^o} e^{-\delta(y-t_{m^o})} dy = \frac{1}{\delta} \Pi^{m^o} = C(t_{m^o}).$$

should hold. Notice that if there is no more imitation after the m^o th imitator, from time t_{m^o} on the benefit of imitation should be less than the cost,

$$\int_{t_{m^o}}^{\infty} \Pi_{t+y}^{m^o+1} e^{-\delta(y-t_{m^o})} dy = \frac{1}{\delta} \Pi^{m^o+1} < C(x) \text{ for all } x \geq t_{m^o}.$$

The condition for optimal imitation date is as follows⁹. Consider the imitation time decision of the i th imitator. Assuming that the number of imitators in the lifetime of the product is more than i , the i th imitator should be indifferent about postponing its date of imitation by a small period of time. So

$$\frac{d(V_{t+t_i}^i(t_i))}{dt_i} - \frac{dC(t_i)}{dt_i} = \Pi_{t+t_i}^i - \rho \lambda \square_\theta e^{-\rho t_i} = 0.$$

Integrating this condition between t_i and t_{i+1} yields the no-profit condition. Therefore once the no-profit conditions hold, the optimality conditions for the imitation time decisions for the imitators will also hold. Using these conditions the equilibrium pattern of imitations can be computed.

⁸In general for the i th imitator, the condition reads

$$\begin{aligned}
\int_{t_i}^{t_{i+1}} \Pi_{t+y}^i e^{-\delta(y-t_i)} dy &= C(t_i) - e^{-\delta(t_{i+1}-t_i)} C(t_{i+1}), \\
&= \left[\lambda \square_\theta e^{-\rho t_i} + (1-\lambda) \square_\theta \right] - e^{-\delta(t_{i+1}-t_i)} \left[\lambda \square_\theta e^{-\rho t_{i+1}} + (1-\lambda) \square_\theta \right].
\end{aligned}$$

⁹Notice that there are m^o unknown imitation dates and $m^o + 1$ equilibrium no-profit conditions for imitation. However these $m^o + 1$ conditions together give the no-profit condition for the innovator. To see that add the left and right hand sides of these conditions to get $V_t^o(0) = \square_\theta$.

Let p_s , m_s , y_s denote the price, the number of producers and the total output of an age- s product in equilibrium. The following proposition states that for $0 < \lambda < 1$ and $\rho > 0$, the number of producers and total output will be increasing whereas the price of the product will be decreasing through time. After some time, though, there will not be imitation anymore and the price and output from then on will be constant. If, on the other hand, $\lambda = 1$, i.e., all of the innovation cost is R&D cost, imitation will not ever stop —hence, output will always be increasing and the price will always be decreasing. The intuition is simple. As long as $\lambda > 0$, the imitation cost will converge to $(1 - \lambda)\square_\theta > 0$ as the product ages, but the benefit from imitation is always strictly decreasing in the number of producers. Therefore there will always be a finite number of imitators, after which imitation will bring strictly negative net profits.

Proposition 1 *Assume that $0 < \lambda < 1$ and $\rho > 0$. In equilibrium, $p_s \searrow p_{s-1}$, $m_s \geq m_{s-1}$, and $y_s \geq y_{s-1}$. There exists an age s^o after which price, number of producers and total output of a product is constant, i.e., $p_j = p^o$, $m_j = m^o$, and $y_j = y^o$ for $j \geq s^o$.*

The fraction of innovation cost that actually goes to R&D, λ , affects the equilibrium innovation and imitation decisions very crucially. The following proposition compares the equilibrium in two economies with different R&D shares in the innovation cost. Consider these two economies at a common number of products. It can be shown that the present value of all future profits from innovating one more product is higher in the low λ economy. Therefore, in equilibrium, number of products in the low λ economy will be higher. In both economies imitation cost will erode through time after innovation and will converge to $(1 - \lambda)\square_\theta$. As the imitation cost converges down to $(1 - \lambda)\square_\theta$, the number of imitators also converges to a fixed number, m^o . Since eventual cost of entry is lower in the high λ economy, m^o should be higher. Therefore it can be shown that eventual number of producers, and eventual total output will be higher, whereas, eventual price will be lower in the high λ economy. Naturally, then, concentration given as the share of the some fixed number of the largest producers in the total industry output will be lower in the high λ economy.

Interestingly, although final number of imitators, and output is high, and final price is low in the high λ economy, there is no monotonicity in this relationship. It can be shown that, up to some time after innovation, number of imitators is actually lower in the high λ economy. This is because higher λ gives more incentives to postpone the imitation dates initially.

Proposition 2 *Consider two economies identical except λ . Let the fraction of innovation cost that goes to research, λ , be higher in the second economy, $\lambda_1 < \lambda_2$. Then, in equilibrium; a) number of goods will be higher in the low λ economy, b) concentration after the industry matures will be lower in the high λ economy, c) $m_1^o \square m_2^o$, $p_1^o \geq p_2^o$, and $y_1^o \square y_2^o$, d) there exists a time x after innovation such that $m_1^t \geq m_2^t$, $y_1^t \geq y_2^t$, and $p_1^t \square p_2^t$ for $0 \square t < x$, whereas, $m_1^t \square m_2^t$, $y_1^t \square y_2^t$, and $p_1^t \geq p_2^t$ for $t > x$.*

Let χ_θ denote the number of type θ product innovations at a point in time. Obviously, if there is innovation in equilibrium for a type θ product the no-profit condition should hold,

$$\frac{\chi_\theta}{\gamma^t} \int_h w_h \lambda(h) dh - V_\theta(0) \begin{cases} > 0 & \text{if } \chi_\theta = 0, \\ \square 0 & \text{if } \chi_\theta > 0. \end{cases} \quad (7)$$

So, the investments needed to create a type θ product will only be undertaken whenever the present value of innovation is non-negative. Since anyone can innovate, in equilibrium, there must be zero rents from doing so.

Licensing innovations is a cost-free activity that can be undertaken by any agent. Since it is a cost-free activity it does not use up any economic resources and since anybody can do it there are no profits from the activity. The agents who manage licenses finance the cost of product development by selling the shares of their product licenses to the people. Each period they pay out profits in dividends. Let the total current dividends and total value of licenses in the economy be denoted by B and Q , respectively. It is clear that the no-arbitrage condition, $1 + r = \frac{Q' + B'}{Q}$, should hold in equilibrium.

4 Equilibrium and Balanced Growth

The aggregate state of the world in this economy is given by, $\mathbf{s} = (z, \phi, \mu)$, where z is the relative productivity of information production, ϕ is the distribution of individuals across states, μ is the distribution of products across types. The equilibrium prices $p_\tau(\theta)$, $f_\tau(\theta)$ and interest rate r , dividend payments and the share price of licenses can all be expressed as a function of the aggregate state of the world, s .

Definition: A competitive equilibrium is a set of allocation rules $\theta(s) = \Theta(s; \mathbf{s})$, $a(s) = A(s; \mathbf{s})$, $\mu_\theta = \Psi_\theta(\mathbf{s})$, $q_\tau(\theta) = \Gamma_\tau(\theta; \mathbf{s})$ together with a set of pricing functions $p_\tau(\theta) = P_{\tau, \theta}(\mathbf{s})$, $r = R(s)$, $f_\tau(\theta) = F_{\tau, \theta}(\mathbf{s})$ such that

1. Agents solve their utility maximization problem ($P1$) with the equilibrium solution to this problem satisfying $\theta(s) = \Theta(s; \mathbf{s})$, $a'(s) = A(s; \mathbf{s})$.
2. Inventors issue the quantity of licenses to maximize their profits in (6), with the equilibrium solution to this problem satisfying $q_\tau(\theta) = \Gamma_\tau(\theta; \mathbf{s})$.
3. Product distribution across types as given by $\mu_\theta = \Psi_\theta(\mathbf{s})$ is determined in accordance with the innovation criteria (7).
4. Markets clear, for the goods market implying

$$\int -\theta(s) \frac{y_\theta(s)}{\mu_\theta} \eta(s) ds = \int d_{\theta\tau}(s; p_\theta, \mathbf{p}) d\phi \text{ for all goods.} \quad (8)$$

where $-\theta(s)$ is an indicator function $-\theta(s) = \begin{cases} 1 & \text{if } \Theta(s; \mathbf{s}) = \theta, \\ 0 & \text{otherwise.} \end{cases}$ and for the financial market implying

$$\int a(s) \eta(s) ds = Q'. \quad (9)$$

where $\eta(s)$ denotes the number of agents with state s . The left hand side in (8) is total output for each good. The first term on the right hand side gives total consumption and the second and third terms give total amount of the good used for product innovation and license renewal respectively.

4.1 Balanced Growth

Along a balanced growth path the amount of output that each individual can produce $y(s)$, the total number of goods, N , the value of licenses Q , and total dividend payments B will all be growing at rate γ .

5 Discussion

In this section we describe some of the properties of the economy with information goods. In the next section we will illustrate these features quantitatively.

Proposition 3 *There will be a surge of innovation of new products in an information age.*

The key feature of information is that it can be reused costlessly. Because of this, the markets for information intensive goods will have the winner-take-all feature. These markets will not be thick, because there will be a smaller and smaller mass of producers producing a product as the product becomes more information intensive. As the technology to produce information improves—as information becomes cheaper and cheaper—more agents will be willing to produce more information intensive products. However because the market becomes less thick as the product gets more information intensive, the economy will create more products. An *information-age* economy will be one that supports a large variety of products—in fact asymptotically as many products as the number of agents. This is in sharp contrast to a mass production economy that would produce a few products with great efficiency. Because of this, traditional productivity measures will be very misleading in an information intensive economy and will not be closely related to welfare. Here we will propose a new measure of well-being that is robust for the information age economy. Those issues will be formally discussed below.

Proposition 4 *Let $\Theta(h)$ denote the product choice of an individual with human capital level h . Then in equilibrium $\Theta(h)$ is increasing in h .*

This proposition states that high human capital agents will choose to produce more information intensive products in equilibrium. The following proposition states that the equilibrium prices for goods and the license fee charged by the inventors are decreasing through time. The license fee charged for information intensive goods are initially higher but eventually lower than the license fee charged for the non-information intensive goods. Also, the second part of the proposition states that information intensive good prices decline more throughout its lifetime.

Proposition 5 *Equilibrium good prices, $p_\tau(\theta)$, and license fees, $f_\tau(\theta)$, are decreasing in τ . Also, $\frac{f_\tau(\bar{\theta})}{f_\tau(\underline{\theta})}$, is monotonically decreasing in τ . Furthermore, the price for the information intensive goods decline more in its lifetime.*

$$\left(\max_{\tau} [p_\tau(\bar{\theta})] - \min_{\tau} [p_\tau(\bar{\theta})] \right) > \left(\max_{\tau} [p_\tau(\underline{\theta})] - \min_{\tau} [p_\tau(\underline{\theta})] \right)$$

As information production becomes more efficient, a larger fraction of the goods will be information intensive. Also, since high human capital agents produce information intensive goods in equilibrium, this improvement in information technology increases the income of high human capital individuals. The effect of this is to increase income inequality.

Proposition 6 *Let $\hat{\mu}(\theta)$ denote the distribution of products over types. Consider two economies identical except $z' > z$. Then, i) $\hat{\mu}(\theta)$ stochastically dominates $\hat{\mu}'(\theta)$; ii) income distribution is more unequal in the primed economy.*

As the information intensity of a good increases, its marginal cost of production decreases. This makes the market for information intensive goods thinner, with fewer producers as the good becomes more information intensive. This is stated in the following proposition.

Proposition 7 *The number of agents producing a type θ good, $\eta(\theta)$, is decreasing in θ .*

Before describing the model economy we use to study the role of information it is useful to describe some of the features of the *new economy* that seem to distinguish it from the recent past. We turn to that discussion in the next section.

5.1 Some Observations

There have been some noteworthy changes in the economic environment in the past two decades that go well beyond the proliferation of computers and the spread of the internet. Many of these have been widely discussed in the press and elsewhere. Here we discuss briefly those that seem most relevant to the properties of the model described above as well as some which follow from those properties. In the following section we analyze the quantitative implications of this model.

- **There has been a surge in product innovation. The “new economy” has been characterized by a dramatic increase in the number of new products.**

Households get utility from the variety of goods and services they can consume. But, determining what makes a good a different variety is somewhat difficult. There are two primary ways of measuring innovation of new goods. One alternative is to use patent data. Patent data has some drawbacks. First, patents don’t represent economically successful innovations, i.e., many patents may not see the final application stage where they are used to produce a new good. On the other hand, there are patents which are successfully used in development of many new goods. Secondly, there can be a relatively long period of time between patenting an innovation and its usage in a new product. Further, patents can be issued for both product innovations and process innovations.

The alternative we follow is to use trademark data.¹⁰ There are some drawbacks to trademark data as well, for example new trademarked goods and services might not really represent different varieties. Nevertheless trademarking a product or a service is a relatively costly and time-consuming process so that a business will not trademark its product unless the probability that the trademark can be used to distinguish the product from the rival products is high enough. The cost of trademarking a product is the same order of magnitude as patenting cost. The other difference

¹⁰According to the definition used by the USA Patent and Trademark Office, a *trademark* “is a word, phrase, symbol, or design, or combination of words, phrases, symbols, or designs which identifies or distinguishes the source of the goods and services of one party from those of others.”

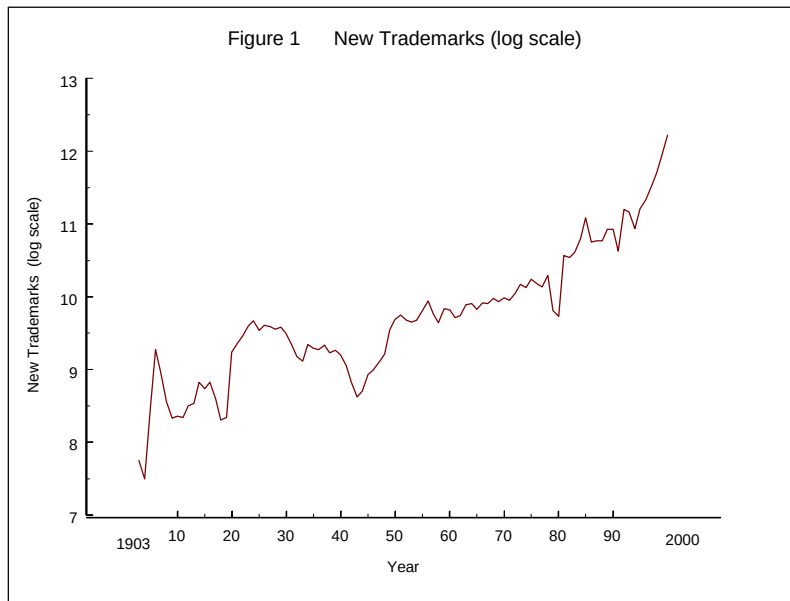


Figure 1: New trademarks in log scale

between patents and trademarks that might be important is that, unlike copyrights and patents, trademark rights can last indefinitely if the owner continues to use the trademark to identify its goods or services and he is willing to pay additional fees. The term of a federal trademark registration is 10 years, with 10-year renewal terms. These renewals are also costly and at the time of renewal some kind of proof that the trademark is actively used is necessary. So, trademark data always tracks economically successful goods or services. That may not be the case for the patent data. Another important advantage of the trademark data is that the average length of time for processing a trademark application is usually no more than ten months.

The number of new trademarks issued between 1903-1997 is shown in Figure 1 on a log scale. The number of new trademarks showed periods of growth both before and after World War I but then declined for nearly twenty years during the Great Depression and World War II. It was relatively level until the 1980's when it began to increase dramatically. One can also look at the stock of trademarks that are active over this period. Although they are not measured directly they can be proxied using

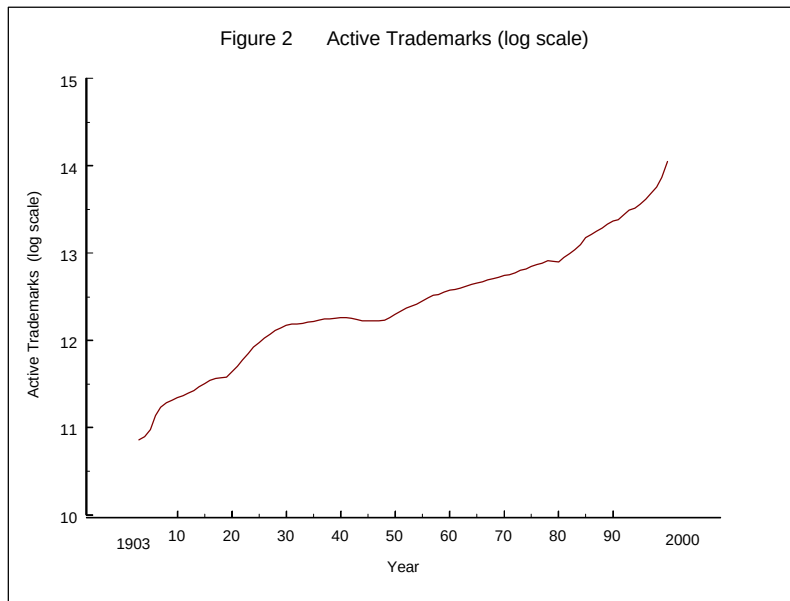


Figure 2: Number of trademarks that are actively used in log scale

the flow of new trademarks issued and a hazard function for existing trademarks. Following these steps Figure 2 plots the estimated stock of active trademarks between 1903-1997 on log scale.

Finally, Figure 3 shows the trend in new trademarks. Again, we can see the dramatic increase starting in the 1970's and 1980's and continuing to the present.

It is worth noting that patent data show the same surge in the recent decades as does the data on the number of new firms entering. Jovanovic and Rousseau (2001) focus attention on the dynamics of firm entry and exit over the past two decades and compare it to the period before and after World War I which is the period of innovation associated with the spread of electrification and the automobile.

- **A rise in income inequality across groups and the growth of winner-take-all markets.**

Beginning in the 1970's and continuing in the 1980's and 1990's inequality of income in the U.S. increased dramatically. Similar patterns have been observed in

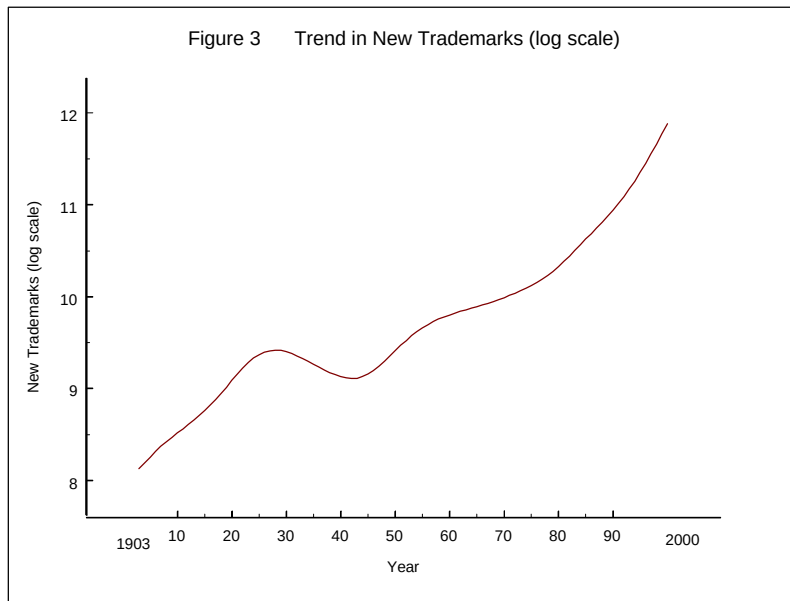


Figure 3: Trend in new trademarks in log scale

other OECD countries. These basic observations have been carefully documented (see e.g. Gottschalk (1997) and Gottschalk and Smeeding (1997) for useful surveys) and do not bear repeating here. Some of this rise in inequality has been due to an increase in the wage premium for skilled workers. Greenwood and Yorukoglu (1997) link this observation to an information technology revolution beginning in 1974 and note that similar patterns were observed in previous industrial revolutions. Many seem to have settled on the explanation that this rise in inequality across skill groups is largely due to skill biased technical change.

Since the basic observations about a rise in inequality are undisputed, we want to focus attention instead on a slightly different interpretation of the rise in inequality that is linked to information. This phenomenon is the growth in winner-take-all markets as has been described for example by Frank and Cook (1995). The notion of winner-take-all markets has been around for a long time. Such markets have been described analytically by Sherwin Rosen (1981) and others. Rosen predicted that new technologies, by increasing the scope of the market for the most talented

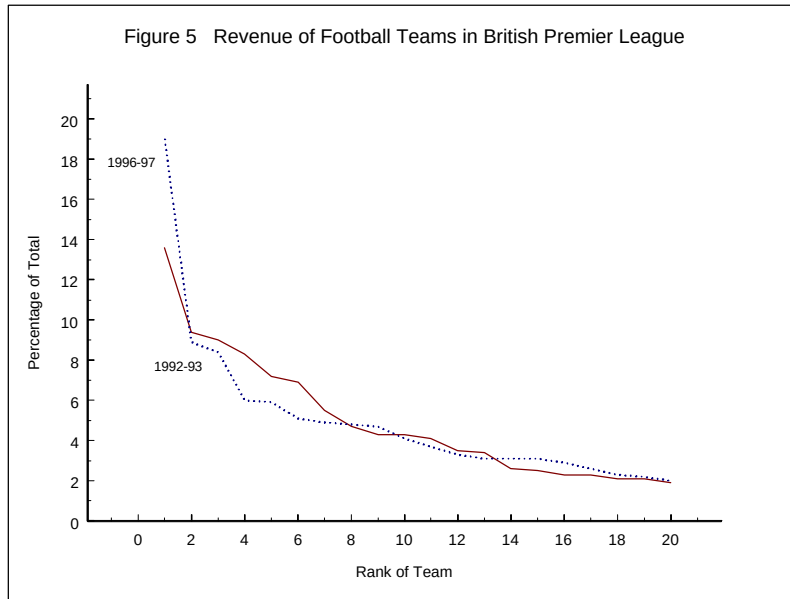


Figure 4: Revenue of football teams in the British Premier League

performers, would increase the inequality of incomes. Frank and Cook argue that this has happened on a broad scale. Their argument is that information inputs allow markets to expand in scope and the rewards for the most successful competitors increase dramatically. Thus, the rise in inequality is directly linked to growth in the information content of goods. There are many examples of this; here we present one drawn from professional sports.

Figure 4 shows the revenues of teams in the British Premier Football League for the period 1992-1993 and 1996-1997. The data are in the accompanying Table. Evident from the figure is that this is an example of market which has become increasingly winner-take-all in that more of the revenues go to the most successful teams. Here the standard deviation of revenues increased from .32 to .39 between the two periods. What changed between 1992-1993 and 1996-1997? The information content of the product changed with the introduction of a sports channel, that made the games of all teams more accessible to the viewing public. This increased the exposure of the most successful teams and resulted in their having a larger share of the revenues. Similar

phenomena are prevalent in other professional sports and in many other markets.

- **A dramatic decline in reproduction and distribution costs of information intensive goods.**

One of the striking changes of recent years is that information intensive goods (like lectures) can be rapidly disseminated to many people at very low cost. There have been decisive changes in the cost structure of media through which information goods are distributed. The following table provides estimates of the share of production, reproduction, and distribution costs as a percentage of total cost for various types of media. By definition these are all high information content goods. What is most remarkable is the dramatic drop in the costs of disseminating information as one moves across media types.

	Production Cost	Reproduction Cost	Distribution Cost	II+III
Type of Media				
Newspapers	20.0	39.5	19.0	58.5
Magazines	29.5	28.1	6.6	34.7
Public service television	55.9	0	9.2	9.2
Commercial television	68.9	0	7.1	7.1
Internet	99	0	1	1

- **Rapidly declining prices of information intensive goods.**

One of the predictions of the model above is that information intensive goods will have steeply declining price profiles. One of the striking features of information intensive goods over the past decade or so has been the rapid decline in their prices. Robert Gordon (2000), among others, has documented the rapid declines in the price of computer hardware and peripherals. He estimates that between 1987 and 1995 prices **declined** at an average four quarter rate of 14.7 percent and from 1996-1999 this increased to a rate of decline of 31.2 %. Software is another good example of

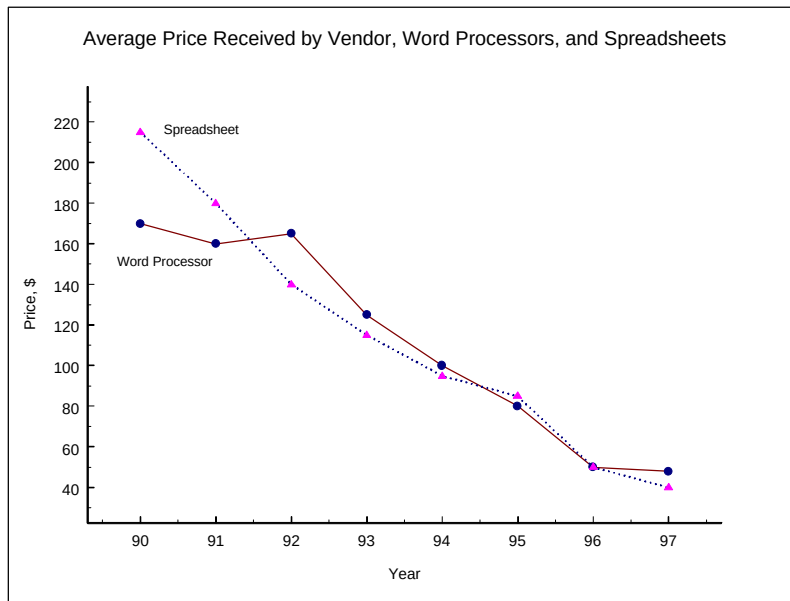


Figure 5: Average price of word processors and spreadsheets.

an even more information intensive good. For most categories of software the price declines very rapidly after its development. The following three graphs illustrate the evolution of prices for some of the most widely used software products - spreadsheets and word processors. These prices are not adjusted for quality improvements or inflation hence they understate the real decline in software prices. The quality improvements in software have been enormous as anyone who experienced DOS versions of WordPerfect of early 90's and recent versions of WordPerfect for Windows would know.

Between 1990-1997 the average rate of price decline for spreadsheets was around 29% a year not adjusted for quality improvements. In the same period the rate of price decline for Word Processors had been around 23% a year.

- **More rapid diffusion of new information intensive products into household's consumption bundles.**

Another prediction of the model described above is that information intensive goods will find their way into consumers consumptions bundles more rapidly because

of their price profiles. This observation is more difficult to document but seems to have substantial support. Table 1 below shows the time elapsed from the date of important innovations to the point at which 25% of households have them as part of their consumption bundles.¹¹

Years To Adoption By 25% of Households			
Invention	Year Invented		Years
Electricity	1873		46
Telephone	1876		35
Automobile	1886		55
Airplane	1903		64
Radio	1906		22
Television	1926		26
VCR	1952		34
Microwave Oven	1953		30
Personal Computer	1975		16
Cellular Phone	1983		13
Internet	1991		7

What is striking in this Table is that time to adoption has declined dramatically for more recent innovations. Many of these, like the VCR, took many years to be commercially available, but once they were adoption was very rapid. Households are able to consume the newest goods more rapidly. The diffusion profiles for a number of goods is show below. It is apparent that these are much steeper for many recent, information intensive, innovations.

This is even more apparent when we compute their diffusion half lives as in the picture below.¹²

¹¹Source: Cox and Alm (1999).

¹²The diffusion half-lives are computed on the basis of the following two regressions

$$T_j = \beta_0 + \beta_d y_j,$$

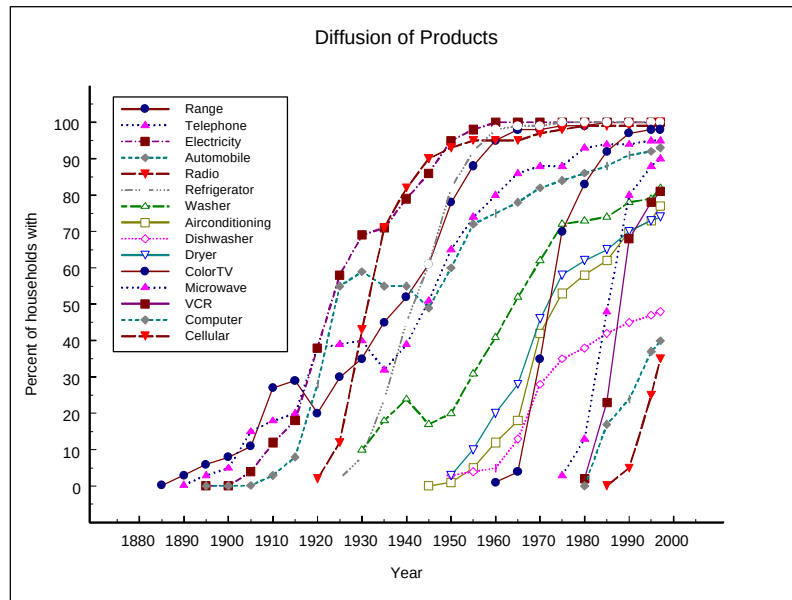


Figure 6: Diffusion curves for major innovations

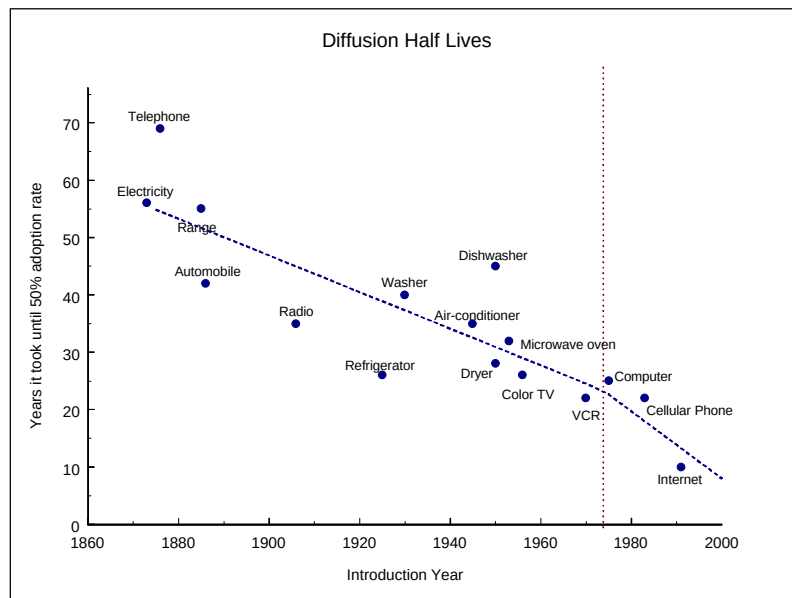


Figure 7: Diffusion half lives for major innovations

All of these observations are consistent with the properties of the model economy described in the previous sections. In the next sections we show the quantitative predictions of the model economy.

6 A Quantitative Example of An Information Revolution

In this section we describe quantitatively what happens as an economy undergoes an information revolution.

6.1 Parameter Values and Functional Forms

The model is calibrated under the assumption that a period in the model is 4 years¹³. Accordingly, the discount factor is $\delta = 1 - 0.97^4 = 0.1147$. The growth rate of productivity and output γ is assumed to be 1.5% a year, i.e., $\gamma = 1.015^4 - 1 = 0.0614$. The elasticity, ϕ , in the utility function is taken to be 0.5.

For computational simplicity we are going to assume that there are two types of products, low-information goods, and high-information goods, rather than a continuum. The elasticities in the production function for high-information and low-information goods are assumed to be 0.1 and 0.9 respectively. Hence $\bar{\theta} = 0.9$ and

and

$$T_j = \beta_0 + \beta_d y_j + \beta_p p_j,$$

where T_j denotes the years it took for j th product to reach 50% of the households starting from the date of introduction, y_j is the year that the product is first introduced and p_j is its price at the date of introduction in time units.

Regression #	β_0	β_d	β_p	R^2
(1)	686.3	-0.336		72.3
t -value		(-6.43)		# of obs = 16
(2)	690.3	-0.338	-0.003	73.1%
t -value		(-5.25)	(-1.85)	# of obs = 14

¹³This length of period is just for illustrative purposes, ten grid points for each time period is used for computation.

$\underline{\theta} = 0.1$.

The initial relative productivity of information input production, z , is calibrated such that at the initial steady state around 20% of the agents produce information intensive goods. The fixed R&D costs for the creation of new high-information and low-information goods are set to $\omega_{\bar{\theta}} = 0.07$ and $\omega_{\underline{\theta}} = 0.02$ respectively. These are determined so that the ratio of R&D cost to production is around 15% for the high-information good and around 5% for the low-information goods. These numbers are roughly consistent with R&D expenditures to sales ratio among firms in high tech sectors like computers, telecommunications and software and in more traditional sectors. Pakes and Schankerman (1978) using patent renewal data find that the patents become obsolete at a rate of 25% a year which suggests a much higher depreciation rate for patents compared to physical types of capital. This can be interpreted as the value of licenses decaying at this rate through time.

Information intensive goods will have R&D costs that are a higher fraction of product development cost. Unfortunately, there is little information about the exact share of direct R&D cost in total development costs. We use $\lambda_{\bar{\theta}} = 0.75$ and $\lambda_{\underline{\theta}} = 0.5$. Let's assume that the cost of imitating the results of existing R&D declines at a rate of 20%. This implies that $\rho = 0.2$.

Empirically, the tail of the income distribution can be well approximated by a Pareto distribution, which is also easy to work with. For that reason the income distribution function $\Lambda(h)$ is represented by the Pareto distribution,

$$\Lambda(h) = 1 - h^{-\sigma}, \text{ for } h \geq 1,$$

with $\sigma = 8$ which approximates U.S. income distribution reasonably well.

6.2 The Steady State

We first describe how this economy behaves in a steady state. In this economy the prices of new products decrease monotonically as they get older because the imitation cost declines through time. Because of the form of the utility function, agents prefer variety but do not prefer any given product over another. Let T denote the age of a product after which no imitation occurs. The lower bound constraint on consumption,

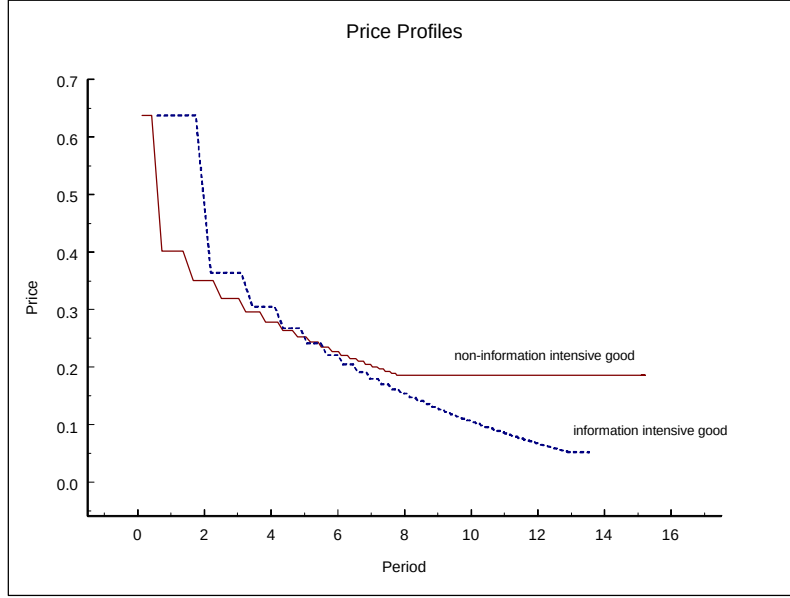


Figure 8: Prices of information and non-information goods through time after innovation

which requires a minimum level of consumption, $\underline{c}$, of each good, prevents most of the agents from consuming all of the goods available. Since, there is no imitation after T periods, equilibrium product prices after age T are all identical. That is $p_{\bar{\theta},i} = p_{\bar{\theta},j}$ for all i, j , and $p_{\underline{\theta},i} = p_{\underline{\theta},j}$ for all i, j . Figure 12 illustrates how the average age of products an agent consumes changes across income levels in the steady state. Those in the lowest percentiles of the income distribution consume a subset of the products that are more than T periods old. Since agents are indifferent about which products they consume we assume they just randomly sample among the old products. In this model economy, agents in the lowest three deciles of the income distribution have the same consumption patterns. Agents above 31st percentile have enough income to buy newer products. Since newer products are more expensive, higher income agents consume newer products on average.

In equilibrium newer goods are more expensive and they include a larger license fee in their price. The license fee paid to the license owner, in turn, covers the product

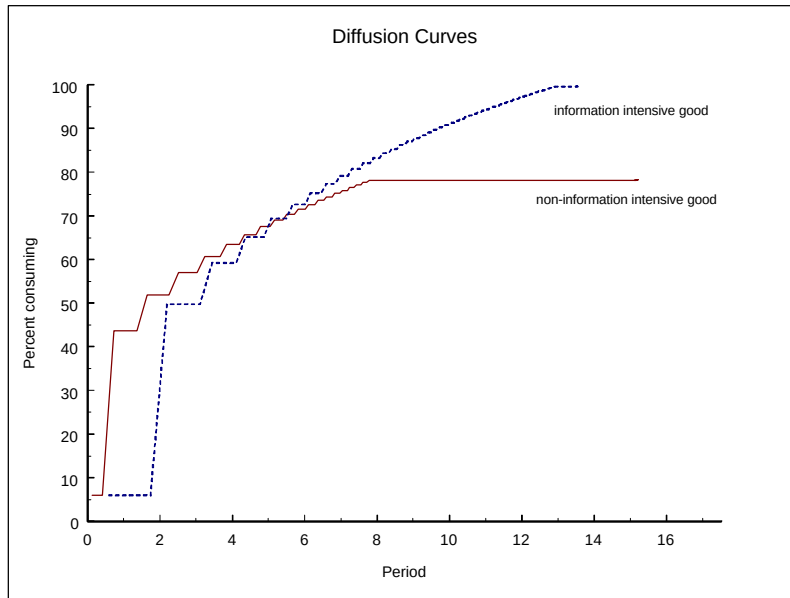


Figure 9: Diffusion curves for information and non-information goods

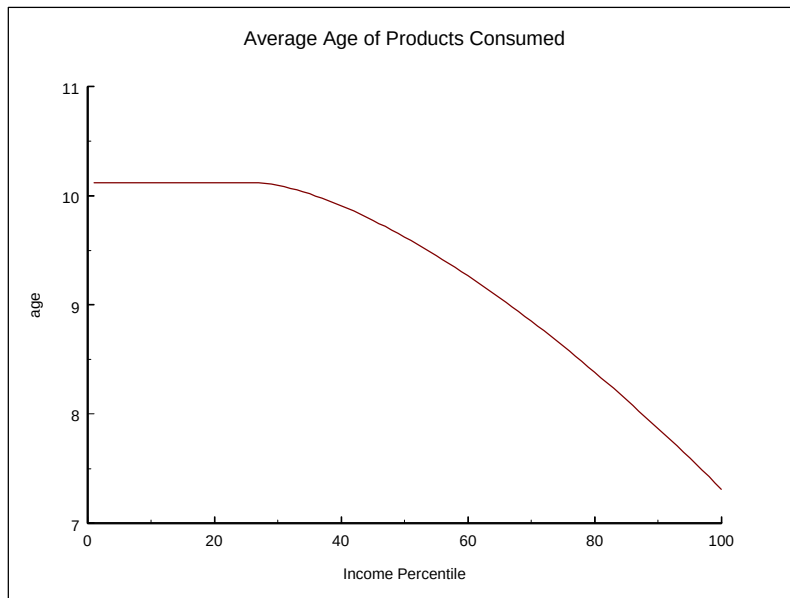


Figure 10: Average age of consumption bundle across income levels

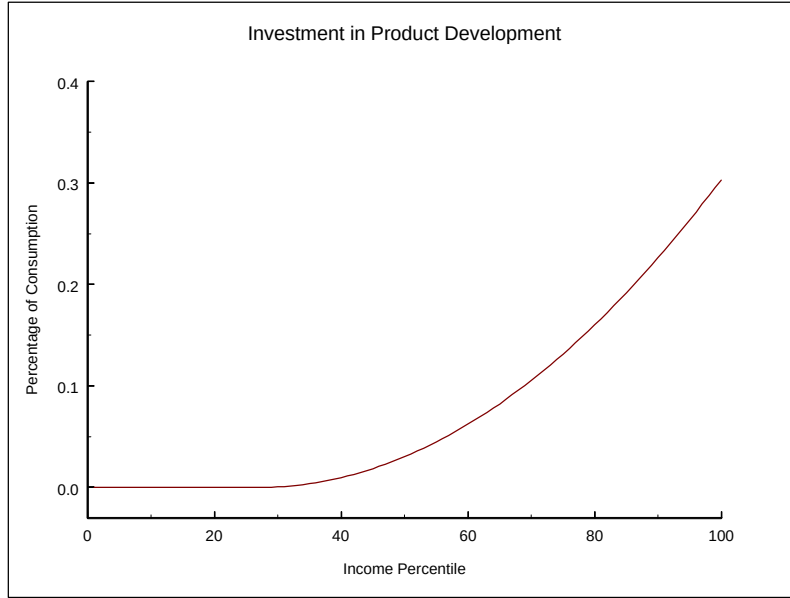


Figure 11: Investment in R&D across income levels

development cost. Agents who consume newer products with higher license fees in them are making the product development investments for the whole society. Figure 13 shows the fraction of total consumption expenditure of agents across income levels that goes to finance the development of new products in the economy. Since those in the lowest percentiles do not consume any products younger than T periods they pay no product development costs. All of their consumption expenditure covers the production cost, with no contribution to product development costs. Agents above the lower percentiles consume newer products and pay higher license fees. In this example around 30% of the total consumption expenditures of the richest 1% goes to pay the product development costs. Since these products eventually raise the utility of every agent this has an aspect of social investment.

6.3 The Information Age

Our main exercise is to consider what happens to an economy that experiences an information technology revolution. Consider an economy in steady state, growing on a

balanced growth path. The production technologies for information input production, $i = z\gamma^t h$, and for non-information input production, $m = \gamma^t n$, are both improving at rate γ . Agents expect the economy to evolve in this fashion indefinitely. Assume, at some date, there is an unexpected breakthrough in the technology of information production and agents have perfect foresight about the future improvements in information that result from this. Assume that the technological breakthrough increases z by 15% a year for 20 years (5 periods). This rate of improvement may seem high, but when one considers the kind of technological improvements information technologies brought, 15% a year is a conservative figure. The German media estimates cited in a previous section suggest around a 50 fold decline in the share of reproduction and distribution costs between newspapers and the internet, and a 9 fold decline between TV broadcasting and the internet. The transitional dynamics of the economy undergoing this change are described next.

Innovation

When the breakthrough in information technology occurs agents realize that providing information-intensive goods to the market will be cheaper in the future. This creates an opportunity to make more profits from the invention of high-information goods. As a result, investment in new good product creation increases dramatically after the breakthrough. This is shown in Figure 15. The ratio of investment in new product development to output rises from an initial level of 5.5% to around 15% . This high level of investment continues for nearly 20 years, after which it falls back to its initial steady state level. These large investments in new product development cause the growth in the number of new products to surge from an initial 1.5% level before the breakthrough to more than 7% after the breakthrough. Again, this boom in new product creation continues for more than 20 years, going back to 1.5% level afterwards. Growth in number of new products is plotted in Figure 14.

Most of the new goods introduced will be information-intensive goods. Figure 16 shows that the fraction of new goods that are information-intensive increases from roughly 20% of new goods to 70% following the breakthrough.

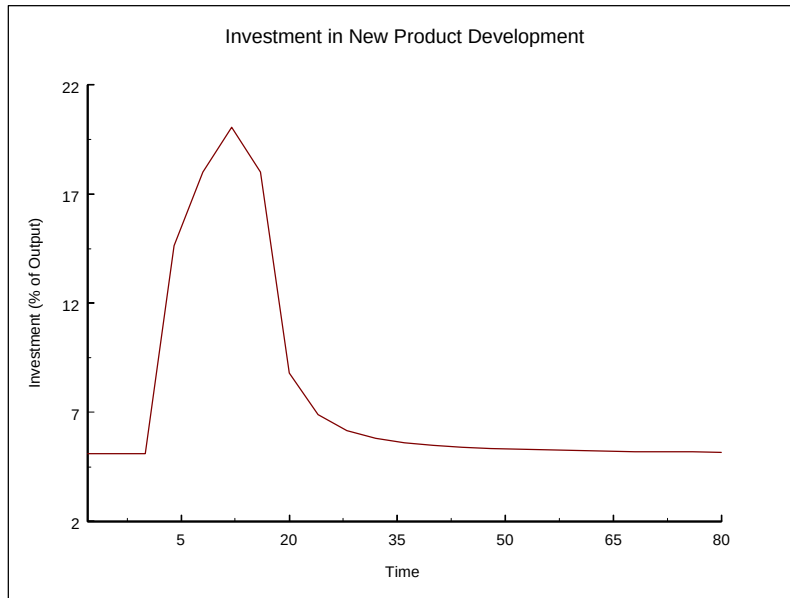


Figure 12: Investment in R&D

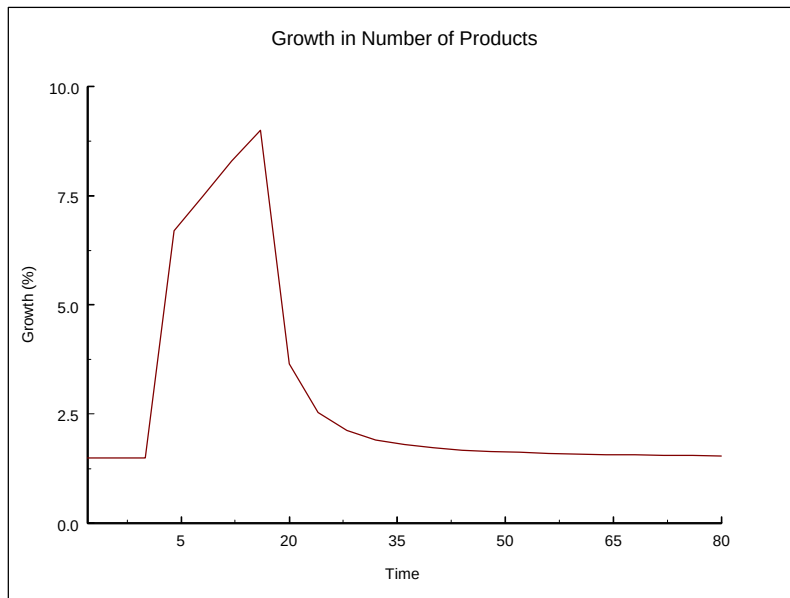


Figure 13: Growth in the number of new products after the breakthrough

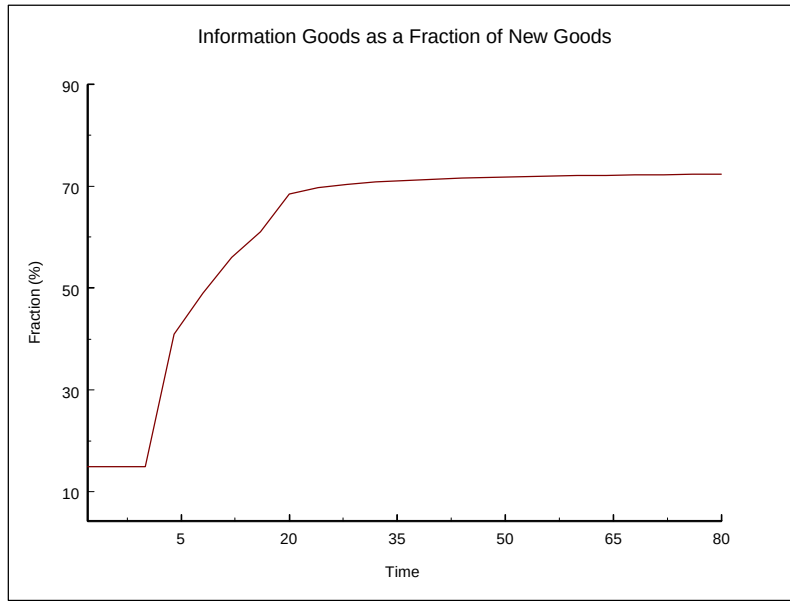


Figure 14: New information goods as a fraction of new goods developed

Productivity

New product development uses economic resources. The increase in the amount of labor allocated to new product development is an investment in future output and welfare. But, the investment in creating new information-intensive goods is not taken into account in conventional productivity measures, leading to an observed slowdown in measured productivity. This slowdown in measured productivity growth continues for almost 20 years. After this temporary slowdown in measured productivity, the cost lowering benefits of the breakthrough in information technology kick in resulting a 20 year period of high growth in measured productivity. Eventually, measured productivity continues to grow at the initial 1.5% a year level. However the gains in measured productivity are permanent. Measured productivity is shown in Figure 17. Greenwood and Yorukoglu (1997) also argue that there will be a slowdown in measured productivity after a technological revolution. In their model learning the new technology takes time and is accomplished by investments made by firms which are not taken into account in productivity estimates.

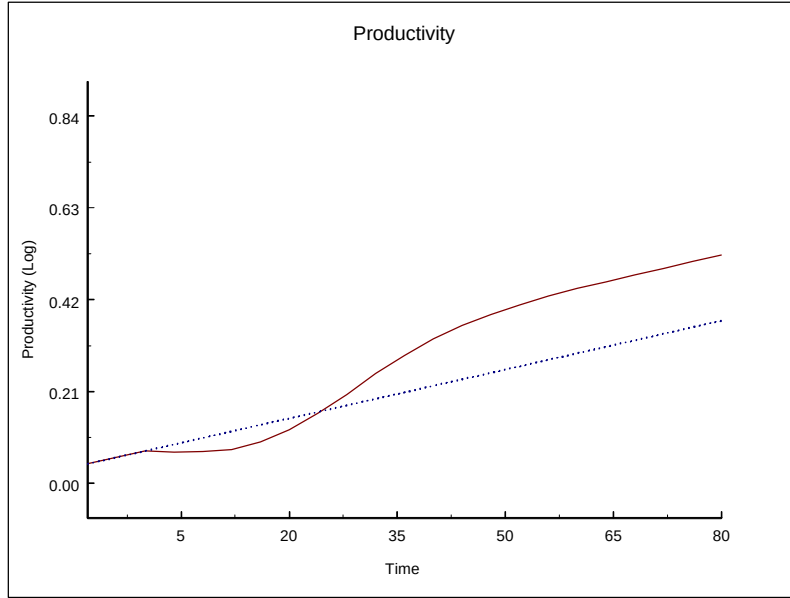


Figure 15: Productivity

Inequality

With the improvements in the distribution technology for high-information goods, producers of these goods, i.e. agents with high human capital become more productive as a group compared to the low human capital agents who produce the low-information goods. Accordingly, the income gap between these two groups will increase. More importantly, among the high-information good producing agents the distribution cost is most binding for the agents with the largest output because the distribution cost function is convex. This means that the efficiency increase in the distribution technology benefits the highest output, highest income, highest human capital agents the most. In other words, the reduction in the information good distribution costs reinforces the winner-take-all nature of the information-intensive goods markets. After the technological breakthrough, the income gap between the rich and the poor widens. Figure 18 plots the logarithm of the ratio of average income of the highest and lowest deciles in the income distribution. By the time 25 years have elapsed, the income gap between these two groups has increased by about 50%.

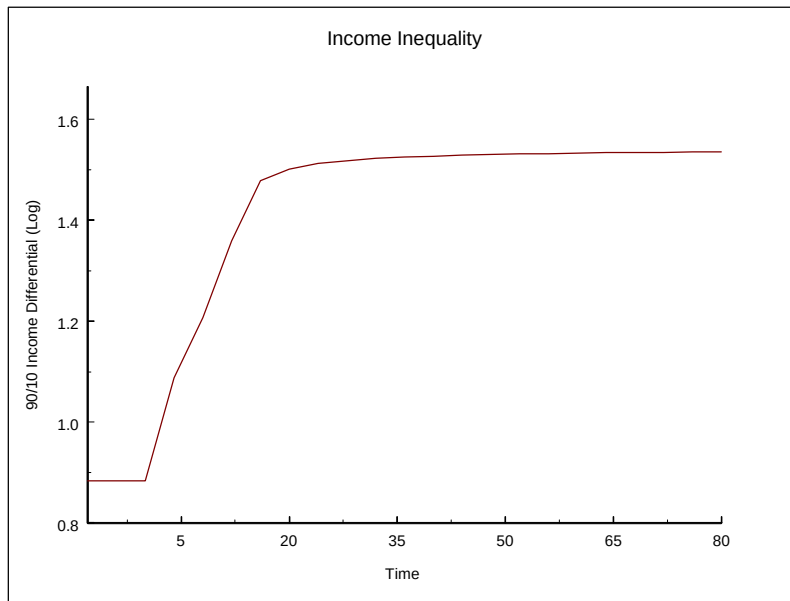


Figure 16: Income inequality

New Product Diffusion

In this economy the important difference between the consumption of the rich and the poor is not that the rich consume more in terms of quantity but that they consume the high price new varieties of goods that the poor cannot afford. Over time as the goods get older the price falls and the income of the poor grows so that they can afford the newer goods. The poor follow the rich in terms of their consumption pattern —the variety of goods consumed— with a certain time lag. The welfare difference between the rich and the poor naturally then depends upon how closely the poor follows the rich. Figure 17 displays the difference in the average age of the consumption goods of the richest 10% and poorest 10%. After the information breakthrough with the boom in new high-information goods, the average age of goods consumed by the rich falls. These new goods are too expensive for the poor. Hence, the differential in the average age of consumption bundles increases for a while. As this wave of new goods get older, the prices fall and the prices of high-information goods fall faster than the prices of low-information goods. As this effect kicks in, the poor begin to catch up to the rich. After a few years the price of high-information

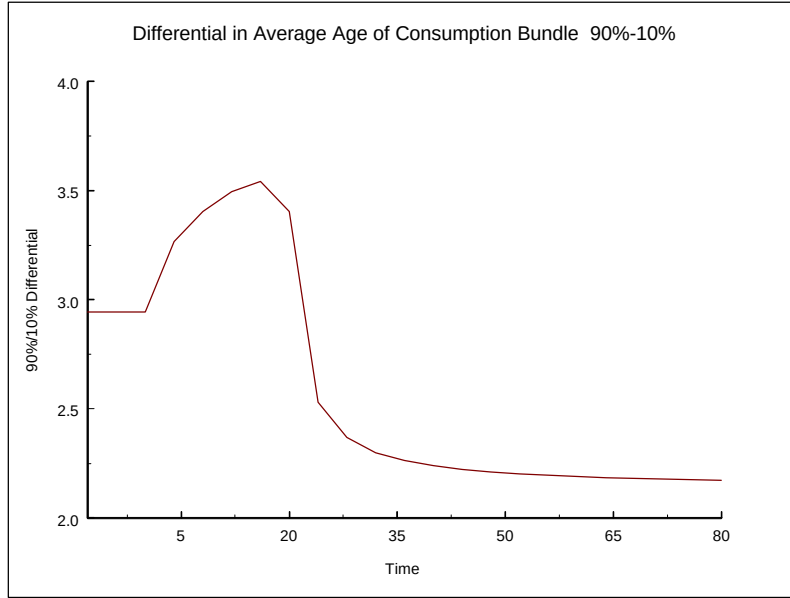


Figure 17: Consumption lag

goods falls sufficiently that the poor can afford them. After the economy converges to the final steady state the gap between rich and the poor is permanently narrowed.

Prices

Although this is not a monetary economy, one can come up with a measure that mimics the consumer price index by the average time cost of producing goods. This is, of course, closely linked to the productivity measure. Figure 18 shows the evolution of this measure. Before the breakthrough, when the economy was evolving at the initial steady state this measure declines at the rate of output growth, $\gamma = 1.5\%$ a year. With the increasing number of new information goods produced after the breakthrough this slows down, even increases for a while (inflation). But as these new information goods age, consumer price index falls sharply, leading to an era of 20 years of strong deflation. After things cool down the economy sinks into the final balanced growth path with an average rate of deflation of $\gamma = 1.5\%$ a year again.

Market Concentration



Figure 18: Consumer price index

With the breakthrough in information technologies, the economy will display higher levels of market concentration. This is mainly due to the near-zero-marginal-cost feature of the high-information goods. The markets for the high-information goods are thin with a small number of producers producing a specific good. Figure 19 plots the logarithm of the average number of producers producing each good through time. This we denote as the *inverse market concentration*. In this economy the number of products is growing over time whereas the number of agents in the economy is constant. Therefore, this measure will be declining through time. Notice, however, that the markets for high-information goods will be relatively more concentrated than those for low-information goods because of the high fixed cost, near zero marginal cost nature of the goods. After the information breakthrough, the boom in the number of new high-information goods increases the average market concentration in the economy dramatically. This can be seen in the Figure 21.

Welfare

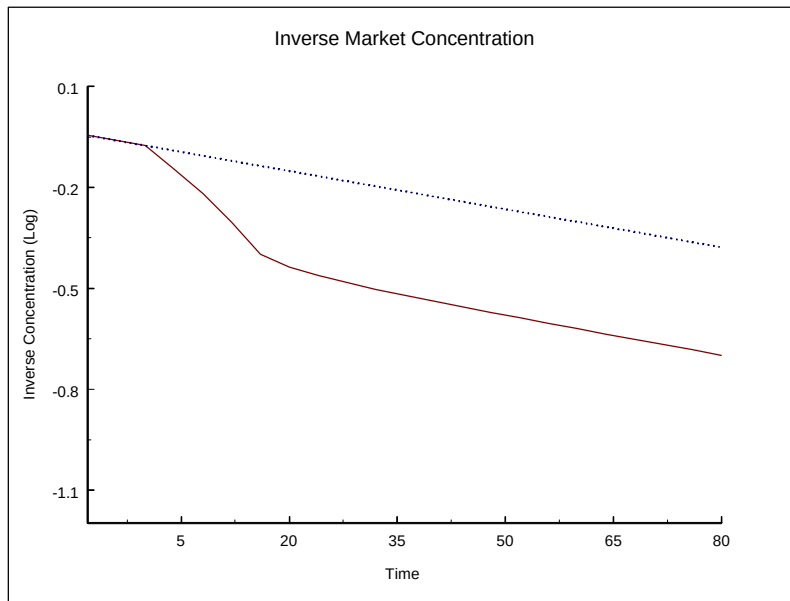


Figure 19: Market concentration

The results so far have shown that an information revolution, by easing the constraints that prevent the more able from capturing a larger market and a bigger share of the pie, increases income inequality. Does this imply that the poor are made worse-off? Figure 22 shows the average welfare gains of the richest 10% and the poorest 10% of the population. The welfare is calculated by figuring how much compensation as a percentage of their income, each group would require to be indifferent between being an agent in the information age at that point and an agent in the extended path of the old economy (with assumed productivity growth continuing). This will measure how much better or worse-off each group of agents are made with the breakthroughs of the information age. The well-being of the rich increases sharply starting from the early days of the information age. There are two channels through which the rich are benefitting early on. First, as producers of information intensive goods, they are becoming more productive with the improvements in the distribution technology. Second, they get to consume these new information intensive toys right away. For those in the lowest decile of the income distribution things do not change very dramatically at the beginning. Their welfare is improving, but the benefits are coming

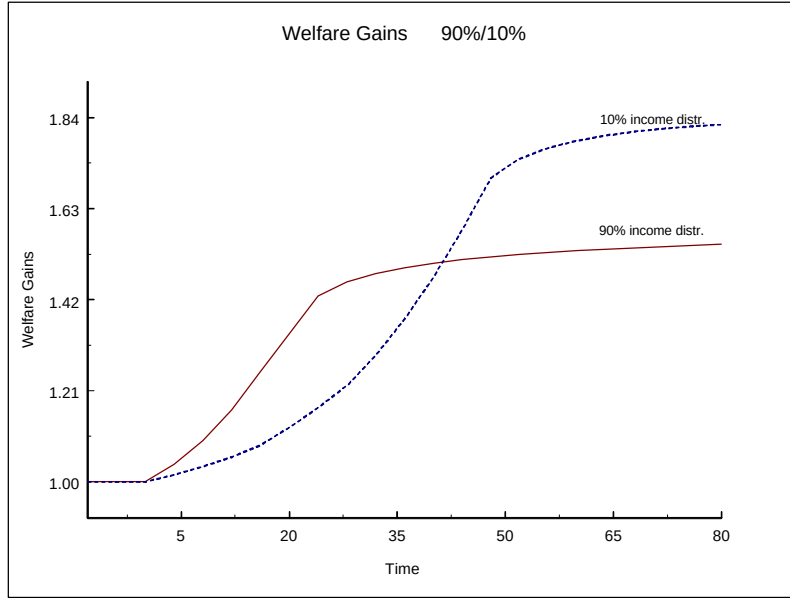


Figure 20: Welfare gains computed in compensating variation sense

slowly. At the beginning the new information intensive goods are expensive and beyond the reach of the poor. The only benefit for the poor during the early years is that the old information goods are becoming cheaper. As time passes, the mass of new information intensive goods produced at the beginning of the information age become cheaper, and the poor start to benefit from them. After almost 20 years from the dawn of the information age, a long era of 30 years begins during which the welfare of the poor increases sharply. The increase is so dramatic that after 40 years or so the welfare gains of the poor outstrip those of the rich. It is important to note that the rise in welfare is due entirely to the fact that those in the upper tail of the income distribution paid the development costs of this shower of new high-information products.

The ratio of the welfare gains of the bottom decile to the top decile is shown in Figure 23. If the poor have keeping-up-with-the-Jones type preferences, they might feel worse-off in the early days of the information age.

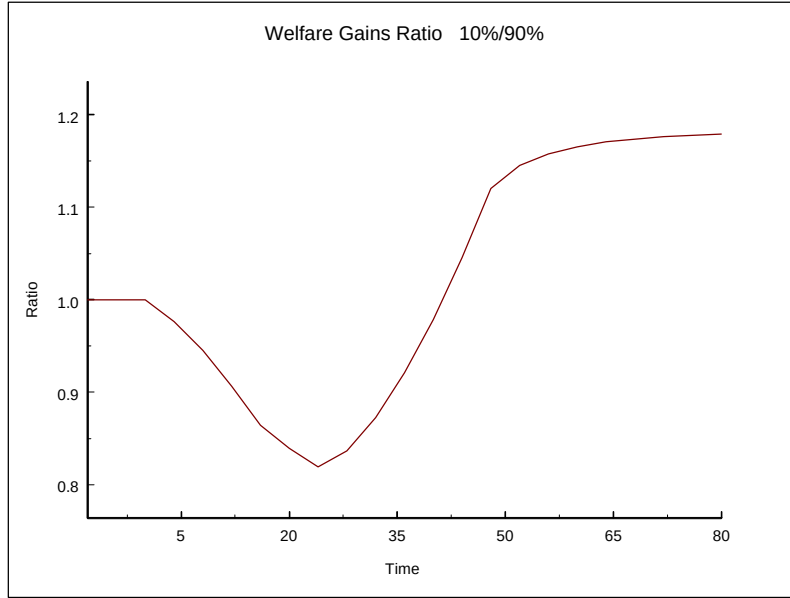


Figure 21: Welfare gains ratio

The Stock Market

The final figure plots the evolution of the total value of licenses which, in this economy, is equivalent to the total value of the stock market. At the date of the technological breakthrough it becomes apparent to agents that the future goods prices and optimal license fees will be declining dramatically. This lowers the expected profits for the low-information good inventions. The value of the low-information good licenses jump down at the date of the breakthrough. However, for high-information goods the increase in efficiency increases expected output and expected license fees of the innovators. As a result, the average value of high-information good licenses jumps up at the date of the breakthrough. Before the breakthrough the low-information good licenses accounted for the larger part of the total license values in the economy. After the breakthrough, the value of high-information good licenses quickly surpass the value of the low-information good licenses. This fall in the value of existing firms and subsequent rise in the value of new firms is exactly the pattern discussed by Greenwood and Jovanovic (1999) and documented further by Jovanovic and Hobijn (2001).

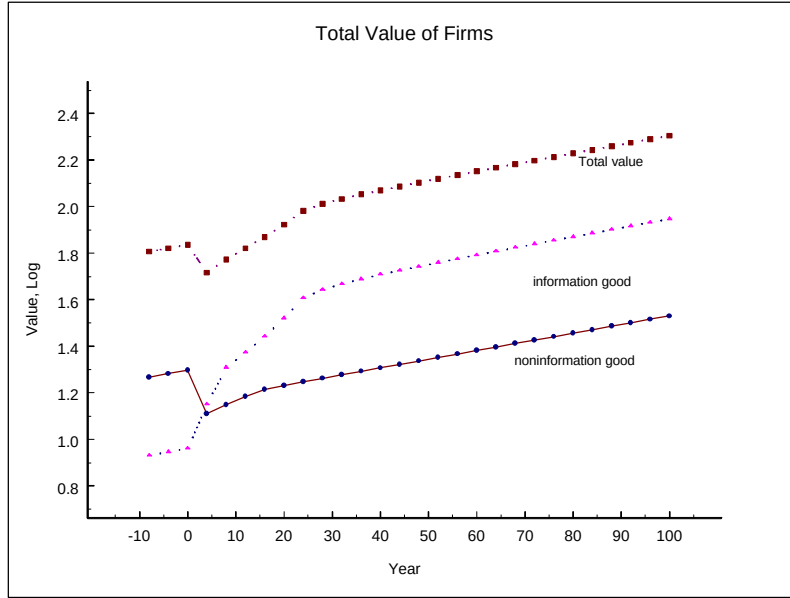


Figure 22: Total value of patents

7 Conclusions

Much has been written about the “New Economy” and the consequences of the information technology revolution. It is clear that, the skeptical views of economists like Robert Gordon (2000) notwithstanding, the characteristics of our economy have changed in important ways. In this paper we treat information formally as an intermediate input to the production process and show that an information revolution, that is an improvement in the production and distribution of information inputs, has dramatic implications for the evolution of the economy that play out over many decades. In particular, an information age will lead to an increase in income inequality, a long fall and then a sharp rise in measured productivity, an increase in concentration, sharp declines in the prices of information intensive goods, a fall and a subsequent increase in the value of the stock market, and an increase in the speed of diffusion of new products to households. All of these features are consistent with observations over the past two and half decades.

In spite of the sharp increase in inequality associated with an information age,

our analysis implies that the welfare of all groups in society will increase with an information revolution precisely because of the increased speed of diffusion of new products throughout the economy. Perhaps even more striking about these findings is the implication that the effects of the information revolution will continue to be felt and influence economic well being for decades to come.

References

- [1] Cairncross, Frances, *The Death of Distance*, Boston: Harvard Business School Press, 2000.
- [2] Cox, W. Michael and Richard Alm, *Myths of Rich and Poor*, New York: Basic Books, 1999.
- [3] Frank, Robert H. and Philip J. Cook, *The Winner Take All Society*, New York: The Free Press, 1996.
- [4] Gordon, Robert J., “Does the New Economy Measure up to the Great Inventions of the Past?” *The Journal of Economic Perspectives* 14 (Fall 2000), 49-74.
- [5] Gottschalk, Peter, “Inequality, Income Growth, and Mobility: The Basic Facts,” *The Journal of Economic Perspectives*, 11 (Spring 1997), 21-40.
- [6] Gottschalk, Peter and Timothy Smeeding, “Cross-National Comparisons of Earnings and Income Inequality,” *Journal of Economic Literature*, 35 (June 1997), 633-87.
- [7] Greenwood, Jeremy and Mehmet Yorukoglu, “1974”, *Carnegie-Rochester Conference Series on Public Policy*, 46 (June 1997), 49-95.
- [8] Greenwood, Jeremy and Boyan Jovanovic, “The Information-Technology Revolution and The Stock Market.” *American Economic Review* (Papers and Proceedings), 89 (May 1999), 116-122.
- [9] Hobijn, Bart, and Boyan Jovanovic, “The IT Revolution and the Stock Market: Evidence,” *American Economic Review*, 2001, 91,5,1203-20.
- [10] Kruger, Dirk and Fabrizio Perri, “Does Income Inequality Lead to Consumption Inequality? Empirical Findings and a Theoretical Explanation,” mimeo, 2000.
- [11] Rosen, Sherwin, “The Economics of Superstars” *American Economic Review*, 71 (Dec. 1981), 845-858.

- [12] Shapiro, and Hal Varian, *Information Rules*, Boston: Harvard Business School Press, 1999.

Appendix

	1992 – 93	1992 – 93	1996 – 97	1996 – 97	Growth in
	Revenue (£m)	%	Revenue (£m)	%	revenue %
Team					
Manchester U.	25,177	13.6	87,939	19.0	249.3
Newcastle U.	8,743	4.7	41,134	8.9	370.5
Liverpool	17,496	9.4	39,153	8.4	123.8
Tottenham H.	16,594	9.0	27,874	6.0	68.0
Arsenal	15,342	8.3	27,158	5.9	77.0
Chelsea	7,891	4.3	23,729	5.1	200.1
Middlesborough	3,968	2.1	22,502	4.9	467.1
Aston Villa	10,175	5.5	22,079	4.8	117.0
Leeds U.	13,324	7.2	21,785	4.7	63.5
Everton	7,994	4.3	18,882	4.1	136.2
Leicester City	4,775	2.6	17,320	3.7	262.7
West Ham U.	6,571	3.5	15,256	3.3	132.2
Notts Forest	7,651	4.1	14,435	3.1	88.7
Sheffiled W.	12,806	6.9	14,335	3.1	11.9
Blackburn R.	6,305	3.4	14,302	3.1	126.8
Sunderland	3,806	2.1	13,415	2.9	252.5
Coventry City	4,592	2.5	12,265	2.6	167.1
Derby County	4,183	2.3	10,738	2.3	156.7
Wimbledon	3,556	1.9	10,410	2.2	192.7
Southampton	4,306	2.3	9,238	2.0	114.5
Total	185,256	100.00	463,949	100.00	150.4